PRACTICE SHEET

1. The multiplicative inverse of $\sqrt{5}+3i$ is

(a) $\frac{\sqrt{5}}{14}+\frac{3i}{14}$ (b) $\frac{\sqrt{5}}{14}-\frac{3i}{14}$

(c) $\frac{\sqrt{5}}{13}-\frac{3i}{13}$ (d) $\frac{\sqrt{5}}{13}+\frac{3i}{13}$

2. If $\left(\frac{1+i}{1-i}\right)^3-\left(\frac{1-i}{1+i}\right)^3 = x + iy$, then (x, y) is equal to

(a) (0, –2) (b) (–2, 0)

(c) (0, 2) (d) (2, 0)

3. If $\frac{(1+i)^2}{2-i} = x+iy$, then the value of x + y is equal to

(a) 5/2 (b) –2/5

(c) 2/5 (d) –5/2

4. The value of $1 + i^2 + i^4 + i^6 + \ldots\ldots i^{2n}$ is

(a) Positive

(b) Negative

(c) Zero

(d) Cannot be determined

5. The value of sum $\sum_{n=1}^{13}\left(i^n+i^{n+1}\right)$ where $i=\sqrt{-1}$, is equal to

(a) i (b) i – 1

(c) –i (d) 0

(c) $\frac{-2}{25}+\frac{i11}{25}$ (d) $\frac{2}{25}+\frac{i11}{25}$

11. If $\frac{z-1}{z+1}$ is a purely imaginary number $(z \neq -1)$, then the value of |z| is

(a) –1 (b) 1

(c) 2 (d) –2

12. If α and β are different complex numbers with $|\beta| = 1$, then $\left|\frac{\beta-\alpha}{1-\bar{\alpha}\beta}\right|$ is equal to

(a) 0 (b) 1/2

(c) 1 (d) 2

13. If $|z + 1| = z + 2(1 + i)$, then the value of z is

(a) $\frac{1}{2}-2i$ (b) $\frac{1}{2}+2i$

(c) $\frac{1}{2}-3i$ (d) $\frac{1}{3}-2i$

14. If z is a complex number, then $(\overline{z}^{-1})(\overline{z})$ is equal to

(a) 1 (b) –1

(c) 0 (d) None of these

15. The complex number z which satisfies the condition $\left|\frac{i+z}{i-z}\right|=1$ lies on